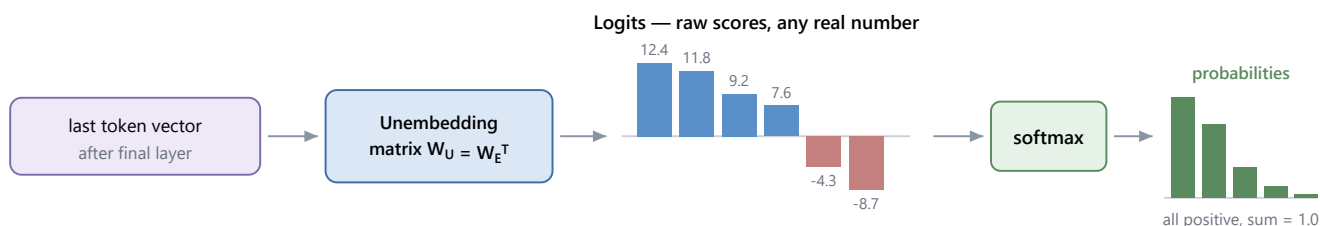


The Language Model Head: Logits and Softmax

After the last layer you have one vector per token. For generating the next token you only need the last one — it is the model's summary of everything so far.

Now we have to get from 5,120 numbers to a score for each of 151,936 possible tokens. That's one more matrix multiply, against the **unembedding matrix** ($151,936 \times 5,120$). Out comes one raw score per token, called a **logit**. Logits can be any number at all — big, small, negative. Softmax then squashes all 151,936 of them into probabilities that are positive and add up to 1.

Weight tying. The embedding matrix and the unembedding matrix have the same dimensions, and they do mirror-image jobs: one turns a token into a vector, the other turns a vector into scores over tokens. So many models simply use *one* table for both, read in the two directions. That saves $151,936 \times 5,120 \approx 778$ **million parameters** — about 5% of this model — for free. It's a real trade, not a free lunch: tying constrains the two jobs to share one set of numbers. Qwen3-14B chooses *not* to tie and keeps two separate 778M tables; the smaller Qwen3 models, where 778M would be a much bigger fraction, do tie. Check a model's config rather than assuming.



And this is exactly where Part 1 picked up: temperature divides the logits before softmax; top-K and top-P filter the list after it; constrained decoding zeroes out anything the schema forbids.

The last vector becomes logits, logits become probabilities, and Part 1's knobs all act right here.

Adding It All Up

We've now named every part. Let's check the arithmetic by rebuilding Qwen3-14B from the pieces — if we understood it, the numbers should come out right.